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KPK BOARD TEXTBOOK NOTES - MATHEMATICS CLASS 9

Unit 11: Parallelograms and Triangles

Complete, easy English notes based on *A Textbook of Mathematics for Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar - the five theorems of the unit (11.1 to 11.5) with Given, To Prove and proof outlines, the mid-segment theorem, the centroid, equal intercepts, worked examples from the book, and exam practice.

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> **How to use these notes:** This is a proof unit. For every theorem, first learn the statement by heart, then the figure, then the proof in the order Given - To Prove - Construction - Proof. In the exam, a theorem answer written in that order, with a neat labelled figure, earns full marks.

What this unit covers

1. Quadrilaterals: parallelogram, rhombus, rectangle, square and trapezoid
2. Theorem 11.1 - opposite sides, opposite angles and diagonals of a parallelogram
3. Theorem 11.2 - when a quadrilateral is a parallelogram
4. Special parallelograms and their diagonals
5. Theorem 11.3 - the triangle mid-segment theorem
6. Theorem 11.4 - medians of a triangle and the centroid (ratio 2 : 1)
7. Theorem 11.5 - equal intercepts on transversals, with its three corollaries
8. Solved exercise highlights, summary, MCQs and practice questions with answers

Source book: KPTB Mathematics for Grade IX, Unit 11 (book pages 242-255). The notes follow the book's theorems, examples and exercises in order. The scan used for verification carries no printed copyright page, so no printing year is claimed here - the 17-unit structure is the current circulating KPTB edition.

1. Quadrilaterals and the Parallelogram Family

A plane figure with three or more straight edges is a **polygon**, named by its number of sides. A polygon with four sides is a **quadrilateral**, and one with three sides is a **triangle**.

Figure	Definition (as given in the book)
Parallelogram	A quadrilateral in which opposite sides are parallel: both pairs of opposite sides are parallel.
Rhombus	A quadrilateral in which all sides are congruent.
Rectangle	A quadrilateral in which all angles are congruent (each angle is 90 degrees).
Square	A quadrilateral in which all sides are congruent and all angles are congruent.
Trapezoid	A quadrilateral with exactly one pair of parallel sides. The parallel sides are called its bases.



In parallelogram ABCD, AB is parallel to DC and AD is parallel to BC. The segments AC and BD are its diagonals.

> **Key point from the book:** a rectangle, a rhombus and a square are all parallelograms, because each of them has both pairs of opposite sides parallel. A square is both a rectangle and a rhombus at the same time.

Words to know before the theorems

- **Congruent** - exactly equal in measure: same length for sides, same degrees for angles.
- **Diagonal** - a segment joining two opposite vertices of a quadrilateral.
- **Bisect** - to divide into two congruent parts. A point that bisects a segment is its **midpoint**.
- **Consecutive angles** - two angles of a quadrilateral that share a side, such as angle A and angle B. Angles with no side in common, such as A and C, are **opposite angles**.
- **Parallel lines** are coplanar lines that do not intersect; **perpendicular lines** intersect to form a right angle.

The sum of the four interior angles of any quadrilateral is 360 degrees (4 right angles).

2. Theorem 11.1 - Properties of a Parallelogram

Theorem 11.1. In a parallelogram: (i) the opposite sides are congruent, (ii) the opposite angles are congruent, and (iii) the diagonals bisect each other.

Given: ABCD is a parallelogram, so AB is parallel to DC and AD is parallel to BC. AC and BD are its diagonals, meeting at E.

To Prove: AB is congruent to DC and AD is congruent to BC; angle A is congruent to angle C and angle B is congruent to angle D; and AE is congruent to CE and DE is congruent to BE (the diagonals bisect each other).

Proof outline (the book's proof, step by step)

Statements	Reasons
AB is parallel to DC and AC cuts them, so angle 1 = angle 2. Similarly angle 3 = angle 4.	Alternate angles
In triangle ABC and triangle CDA: AC is common, angle 3 = angle 4, angle 1 = angle 2. So triangle ABC is congruent to triangle CDA.	ASA congruence
Therefore AB = CD and BC = DA, and angle B = angle D.	Corresponding parts of congruent triangles
Angle 1 + angle 4 = angle 2 + angle 3, so angle A = angle C.	Adding the equal alternate angles
In triangle ABE and triangle CDE: AB = CD (proved), angle 1 = angle 2, and angle AEB = angle CED. So the two triangles are congruent.	SAA congruence; vertical angles
Therefore AE = CE and BE = DE.	Corresponding parts - the diagonals bisect each other

> **Corollary (from the book):** A diagonal of a parallelogram divides the parallelogram into two congruent triangles. In the figure above, diagonal AC gives triangle ABC congruent to triangle CDA - this single fact is the engine of the whole proof.

One more property to remember

The book's Remember box adds a fourth result: in a parallelogram, **consecutive angles are supplementary** - they add to 180 degrees. So if angle A is 70 degrees, the angles next to it, B and D, are each 180 - 70 = 110 degrees.

3. Using Theorem 11.1 - Worked Examples

Almost every numerical question in this unit uses one rule: **set the opposite parts equal**, or **add consecutive angles to 180 degrees**.

Example 1 (the book's example). Quadrilateral WXYZ is a parallelogram. Side XY is labelled x - 9, the opposite side WZ is 15, angle W is 60 degrees and the opposite angle Y is labelled y degrees. Find x and y.

Solution: Opposite sides are equal, so XY = WZ: x - 9 = 15, giving **x = 24**. Opposite angles are equal, so **y = 60**.

Example 2. One angle of a parallelogram is 70 degrees. Find the remaining angles.

Solution: The angle opposite the 70-degree angle is also 70 degrees. Each consecutive angle is 180 - 70 = 110 degrees. The four angles are **70, 110, 70 and 110 degrees**. Check: 70 + 110 + 70 + 110 = 360 degrees.

Example 3. The diagonals of parallelogram ABCD meet at E. If AC = 14 cm and BD = 10 cm, find AE and BE.

Solution: The diagonals bisect each other, so E is the midpoint of both. AE = 14 / 2 = **7 cm** and BE = 10 / 2 = **5 cm**.

Example 4. In parallelogram PQRS, side PQ = 3x - 5 and the opposite side SR = 25. Find x and the perimeter if PS = 12.

Solution: Opposite sides are equal: 3x - 5 = 25, so 3x = 30 and **x = 10**. The other pair of sides is PS = QR = 12, so the perimeter is 2 x (25 + 12) = **74 units**.

Watch out: the diagonals of a general parallelogram bisect each other, but they are **not** equal in length and **not** perpendicular. Equal diagonals belong to the rectangle, and perpendicular diagonals belong to the rhombus - see page 6.

4. Theorem 11.2 - A Test for a Parallelogram

Theorem 11.1 starts with a parallelogram and finds its properties. Theorem 11.2 works in the opposite direction: it tells us how to *prove* that a quadrilateral is a parallelogram.

Theorem 11.2. If two opposite sides of a quadrilateral are congruent and parallel, then the quadrilateral is a parallelogram.

Given: ABCD is a quadrilateral in which AB is congruent to DC and AB is parallel to DC.
Construction: Join A to C.

Proof outline

Statements	Reasons
In triangle ABC and triangle CDA: AB = DC, AC is common, and angle 1 = angle 2. So triangle ABC is congruent to triangle CDA.	Given; common side; alternate angles; SAS congruence
Therefore angle 3 = angle 4.	Corresponding angles of congruent triangles
Angles 3 and 4 are alternate angles for lines AD and BC cut by AC, so AD is parallel to BC.	Equal alternate angles give parallel lines
ABCD now has both pairs of opposite sides parallel, so ABCD is a parallelogram.	Definition of a parallelogram

Other tests worth knowing

- **Both pairs of opposite sides congruent:** the book's summary states that if two pairs of opposite sides of a quadrilateral are congruent, then the quadrilateral is a parallelogram.
- **Midpoints of a quadrilateral:** the line segments joining the midpoints of the sides of any quadrilateral, taken in order, form a parallelogram (Review Exercise 11, Question 4). The proof joins a diagonal and applies the mid-segment theorem, Theorem 11.3, in the two triangles it makes.

> **Do not mix the directions up.** "Opposite sides of a parallelogram are equal" (11.1) uses a figure already known to be a parallelogram. "One pair of opposite sides equal and parallel" (11.2) is used to prove that a figure *is* a parallelogram. Both conditions - congruent *and* parallel, on the *same* pair of sides - are needed in 11.2.

5. Special Parallelograms: Rectangle, Rhombus, Square

Because the rectangle, rhombus and square are parallelograms, everything in Theorem 11.1 is true for them. Each one then adds a special property of its own, and the exam tests these through the diagonals.

Figure	Extra property	Diagonals
Parallelogram	Opposite sides and angles congruent; consecutive angles supplementary	Bisect each other
Rectangle	All angles 90 degrees	Bisect each other **and are congruent**
Rhombus	All sides congruent	Bisect each other **at right angles, and bisect the vertex angles**
Square	All sides congruent and all angles 90 degrees	Bisect each other, **congruent and perpendicular**

Two proofs the book asks for

- **Diagonals of a rhombus bisect its angles** (Exercise 11.1, Question 6). A diagonal splits the rhombus into two congruent isosceles triangles (SSS), so the two parts of each vertex angle are equal.
- **Diagonals of a rectangle are congruent** (Exercise 11.1, Question 8). The two triangles formed by the diagonals with a common side are congruent by SAS, using the right angles and the equal opposite sides.

Example. In rhombus MNOP, angle N is 70 degrees. What is angle O?

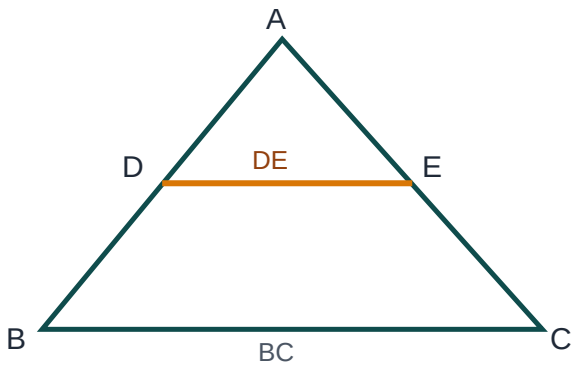
Solution: A rhombus is a parallelogram, and O is consecutive to N, so angle O = 180 - 70 = **110 degrees**. (If the question had asked for angle P, opposite to N, the answer would be 70 degrees.)

MCQ trap: "Diagonals are perpendicular" is true for a rhombus *and* a square. "Diagonals are congruent *and* perpendicular" can only be the **square** - that is Review Exercise 11, MCQ (i).

6. Theorem 11.3 - The Mid-Segment Theorem

A **mid-segment** of a triangle is a segment that connects the midpoints of two sides of the triangle. Every triangle has three mid-segments.

Theorem 11.3 (Mid-Segment Theorem). The line segment joining the midpoints of two sides of a triangle is parallel to the third side and is equal to one half of its length.



D and E are the midpoints of AB and AC. Then DE is parallel to BC and DE = 1/2 BC.

Given: In triangle ABC, D and E are the midpoints of AB and AC, and DE joins them. **To Prove:** DE is parallel to BC and DE = 1/2 BC.

Proof idea (the book's construction)

On DE produced, take a point F such that DE = EF, and join F to C. Triangles ADE and CFE are congruent by SAS (DE = EF by construction, AE = CE because E is a midpoint, and the vertical angles at E are equal). This gives AD = CF and the alternate angles that make AB parallel to CF. Since AD = DB, we get DB = CF with DB parallel to CF, so BCFD is a parallelogram (Theorem 11.2). Therefore DE is parallel to BC, and DF = BC - but DE is half of DF, so DE = 1/2 BC.

Example (from Exercise 11.1, Question 5). DE is a mid-segment of triangle ABC. (i) If BC = 26, find DE. (ii) If DE = 5 and DE is parallel to AB, find AB. (iii) If AB = 6 and DE is its mid-segment, find DE.

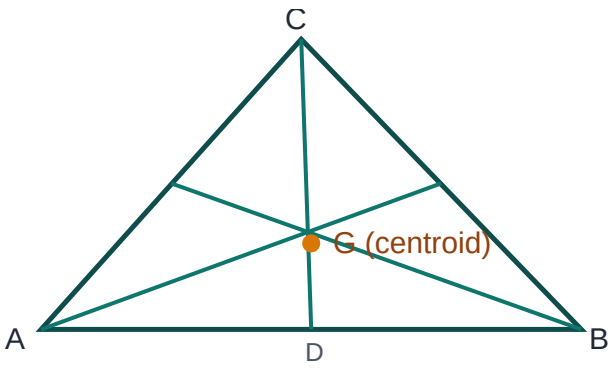
Solution: (i) DE = 26 / 2 = **13**. (ii) AB = 2 x 5 = **10**. (iii) DE = 6 / 2 = **3**. A mid-segment is always half of the side it is parallel to - never the other way round.

7. Medians and the Centroid - Theorem 11.4

A **median** of a triangle is a line segment from a vertex to the midpoint of the opposite side. A triangle has three medians, and remarkably they all pass through one single point.

Theorem 11.4. The medians of a triangle are concurrent, and their point of concurrency is the point of trisection of each median.

The point where the three medians meet is called the **centroid** of the triangle. "Point of trisection" means the centroid divides each median into two parts in the ratio **2 : 1** - the part towards the vertex is twice as long as the part towards the side. In the book's notation, for median AE with centroid G: GE = 1/2 AG, that is, AG : GE = 2 : 1 and AG = 2/3 of the whole median.



The three medians meet at the centroid G. On each median, the vertex part is double the side part.

Proof idea

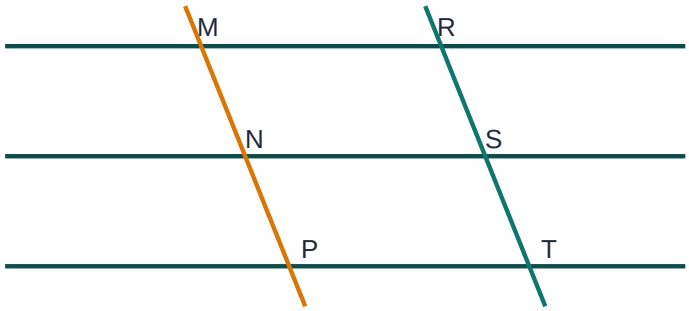
Take two medians meeting at G. Double the segment CG to a point H and join H to A and B. The mid-segment theorem (11.3), used in triangles ACH and BCH, shows AHBG is a parallelogram, whose diagonals bisect each other - so the third line through G is also a median, and G cuts each median in the ratio 2 : 1.

Example (Exercise 11.1, Question 10). G is the centroid of triangle ABC, with BG = 6, AF = 12 and AE = 15, where BF and AE are medians and F is the midpoint of AC. Find FC, BF, AG and GE.

Solution: F is a midpoint, so FC = AF = **12**. On median BF, BG : GF = 2 : 1, so GF = 6 / 2 = 3 and BF = 6 + 3 = **9**. On median AE, AG = 2/3 x 15 = **10** and GE = 15 - 10 = **5**.

8. Theorem 11.5 - Equal Intercepts on Transversals

Theorem 11.5. If three or more parallel lines make congruent intercepts on a transversal, they also intercept congruent segments on any other line that cuts them.



Three parallel lines cut transversal LX at M, N, P with MN = NP. Then on any other transversal they cut RS = ST.

Proof idea: Through R and S draw lines parallel to the first transversal. These make two small parallelograms, which transfer the equal intercepts MN and NP across to the second transversal as equal segments RU and SV. Triangles RUS and SVT are then congruent by SAA, giving RS = ST.

The three corollaries - learn these, they are asked directly

1. **Corollary 1:** If a line is drawn from the midpoint of one side of a triangle and parallel to another side, it will bisect the third side. (This is the converse direction of the mid-segment theorem.)
2. **Corollary 2:** If a line is drawn parallel from the midpoint of a non-parallel side of a trapezoid to the parallel sides, it will bisect the other non-parallel side.
3. **Corollary 3:** If one side of a triangle is divided into congruent segments, the lines drawn from the points of division parallel to another side will make congruent segments on the third side.

Example. Three parallel lines are cut by a transversal at P, Q, R with PQ = QR = 4 cm. A second transversal cuts the same lines at X, Y, Z, and XY = 4 cm. What is YZ and XZ?

Solution: By Theorem 11.5, YZ = XY = **4 cm**, so XZ = 4 + 4 = **8 cm**. Equal spacing on one transversal always means equal spacing on the other.

9. Solved Exercise Highlights (Book Exercise 11.1 and Review)

Question	Answer
One angle of a parallelogram is 70 degrees. Find the remaining angles. (Ex 11.1 Q1)	110, 70 and 110 degrees
One exterior angle of a parallelogram is 125 degrees. Find all interior angles. (Ex 11.1 Q2)	Interior angle next to it: 180 - 125 = 55 degrees; angles are 55, 125, 55, 125 degrees
RSTU is a parallelogram with RU = 3n - 20 and ST = n + 12. Find ST. (Ex 11.1 Q3)	Opposite sides equal: 3n - 20 = n + 12, so n = 16 and ST = 28
Find each variable (Ex 11.1 Q4): (i) x opposite 9, y opposite 15; (ii) m + 1 opposite 6, n opposite 12; (iii) angle 2p opposite 120 degrees; (iv) angle (d - 21) opposite 105 degrees, side z - 8 opposite 20.	(i) x = 9, y = 15 (ii) m = 5, n = 12 (iii) p = 60 (iv) d = 126, z = 28
One angle of a parallelogram is 60 degrees. Find the remaining angles. (Review Q2)	120, 60 and 120 degrees
One exterior angle of a parallelogram is 130 degrees. Find all interior angles. (Review Q3)	50, 130, 50 and 130 degrees
In rhombus MNOP, angle N = 70 degrees. Find angle O. (Review Q5)	110 degrees (consecutive angles)
One angle of a parallelogram is 50 degrees. Find the remaining angles. (Review Q6)	130, 50 and 130 degrees

> **The pattern to notice:** each question above is solved by one line of Theorem 11.1 - opposite parts equal, or consecutive angles adding to 180 degrees. For an exterior angle, first use interior + exterior = 180 degrees.

Proof questions set in the book - know the plan for each

- Diagonals of a rhombus bisect its angles - SSS on the two triangles made by a diagonal.
- Diagonals of a rectangle are congruent - SAS using the right angles.
- Joining the midpoints of a quadrilateral's sides, in order, gives a parallelogram - join one diagonal and use Theorem 11.3 in both triangles.
- Joining the midpoints of a triangle's three sides makes four congruent triangles - each mid-segment is half of, and parallel to, a side.

10. Unit Summary and Common Mistakes

The whole unit on one page

- **Parallelogram:** both pairs of opposite sides parallel. Rectangle, rhombus and square are all parallelograms.
- **11.1:** Opposite sides congruent, opposite angles congruent, diagonals bisect each other; consecutive angles supplementary; a diagonal makes two congruent triangles.
- **11.2:** One pair of opposite sides congruent *and* parallel proves a parallelogram. So do two pairs of opposite sides congruent.
- **Diagonals:** rectangle - congruent; rhombus - perpendicular and bisecting the angles; square - congruent and perpendicular.
- **11.3:** Mid-segment is parallel to the third side and half of it: DE || BC and DE = 1/2 BC.
- **11.4:** Medians meet at the centroid, which divides each median in the ratio 2 : 1 from the vertex.
- **11.5:** Equal intercepts on one transversal mean equal intercepts on every transversal; a line through a midpoint parallel to one side bisects the next side.
- **Angles:** the four interior angles of a quadrilateral add to 360 degrees.

Common mistakes

- **Opposite vs consecutive:** opposite angles are *equal*; consecutive angles *add to 180 degrees*. Most lost marks in this unit come from swapping these two.
- **Half the wrong way:** the mid-segment is half of the third side, so the third side is *double* the mid-segment - check which one the question gives you.
- **Centroid ratio reversed:** the longer part (2 parts) is always on the *vertex* side of the centroid, never on the side towards the midpoint.
- **Assuming special diagonals:** a plain parallelogram's diagonals are neither equal nor perpendicular. Only claim that for a rectangle, rhombus or square.
- **Theorem 11.2 needs both conditions:** one pair of sides equal alone, or parallel alone, does not prove a parallelogram.
- **Proof presentation:** writing statements without reasons, or skipping Given / To Prove / Construction, loses marks even when the idea is right.

11. Practice - MCQs and Questions (Book Review Exercise 11)

1. Which quadrilateral must have diagonals that are congruent and perpendicular? (a) Rhombus (b) Square (c) Trapezoid (d) Parallelogram
2. How many equilateral triangles are made by joining the midpoints of an equilateral triangle? (a) 2 (b) 3 (c) 4 (d) Cannot be determined
3. Which statement must be true? (a) If a parallelogram is not a rectangle, it is not a square (b) If a parallelogram is not a square, it is not a rectangle (c) All rectangles are squares (d) If a parallelogram is a rectangle, it is a square
4. Which quadrilateral's diagonals are perpendicular and bisect the opposite angles? (a) Trapezoid (b) Rectangle (c) Rhombus (d) Parallelogram
5. If opposite angles of a quadrilateral are equal and none is a right angle, it is a: (a) Square (b) Parallelogram (c) Trapezoid (d) Rectangle
6. Medians of a triangle are divided by the point of concurrency in the ratio: (a) 2 : 1 (b) 2 : 3 (c) 1 : 3 (d) None of these
7. The centroid is the point of concurrency of the: (a) Medians of a triangle (b) Angle bisectors (c) Altitudes (d) Perpendicular bisectors
8. If angle A + angle C of parallelogram ABCD is 120 degrees, then angle B is: (a) 25 degrees (b) 50 degrees (c) 65 degrees (d) None of these
9. Diagonals of a square are ..... to each other. (a) Perpendicular (b) Not congruent (c) Parallel (d) None of these
10. The sum of the measures of the interior angles of a quadrilateral is: (a) 2 right angles (b) 4 right angles (c) 3 right angles (d) None of these

Short questions

1. One angle of a parallelogram is 80 degrees. Find the other three angles.
2. In triangle PQR, S and T are the midpoints of PQ and PR. If QR = 18 cm, find ST, and state what else is true about ST.
3. A median of a triangle is 21 cm long. How far is the centroid from the vertex, and from the midpoint of the side?
4. State the mid-segment theorem, with its Given and To Prove.

> **Answer key - MCQs:** 1 (b) 2 (c) 3 (a) 4 (c) 5 (b) 6 (a) 7 (a) 8 (d) - angle A = angle C = 60 degrees, so angle B = 120 degrees, which is not listed 9 (a) - the diagonals of a square are also congruent 10 (b).
> **Short questions:** 1. 100, 80 and 100 degrees. 2. ST = 9 cm, parallel to QR. 3. 14 cm from the vertex, 7 cm from the midpoint. 4. Theorem 11.3.